Validation Report

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Report Content

SRS VALIDATION REPORT
Prepared in accordance with IEEE 830-2013 and ISO/IEC/IEEE 29148:2018 Standards

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1. Executive Summary

The Software Requirements Specification (SRS) for *Archivist v1.1* demonstrates strong foundational quality, exhibiting commendable structure, clarity, and alignment with industry standards. It comprehensively covers core functional and non-functional domains, employs consistent terminology, and includes well-articulated acceptance criteria and traceability tags. However, several gaps—particularly in **structural completeness**, **cross-cutting consistency**, and **operational verifiability**—prevent the document from achieving full readiness for downstream development and certification.

The SRS **passes validation with conditions**: it is technically sound and largely compliant, but requires targeted revisions to satisfy mandatory conformance criteria (especially IEEE 830 §4.3.8 and ISO/IEC/IEEE 29148 §4.3.8 on traceability), resolve latent contradictions, and strengthen feasibility grounding.

Overall confidence level: **High (85%)**, pending resolution of Major and Critical items below.

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2. Issues Found (Categorized by Severity)

Severity	Description	Standard Reference(s)	Impact
Critical	**Missing explicit Traceability Matrix** — No dedicated matrix mapping requirements to stakeholder needs, source documents, test cases, or design elements. This violates mandatory traceability provisions in both IEEE 830 §4.3.8 and ISO/IEC/IEEE 29148 §4.3.8.	IEEE 830 §4.3.8; ISO/IEC/IEEE 29148 §4.3.8	Blocks formal verification, audit readiness, and regulatory compliance (e.g., ADA, FERPA, HIPAA-aligned data governance).
Major	**Inconsistent Role-Based Access Logic** — Contradictory scope definitions between FP-007 (HR sees *only* ADA accommodations) and FR-006 (RBAC "at all layers" without clarifying HR's *full* data scope). Ambiguity risks policy non-compliance.	ISO/IEC/IEEE 29148 §5.2.2 (Consistency); §6.3 (Security)	High risk of security misconfiguration, access control violations, or audit failure. Requires authoritative role-definition reconciliation.

Severity	Description	Standard Reference(s)	Impact
Major	**Unspecified Performance Thresholds & Verification Methods** — PR-003 (1-sec GPS recalculation), FR-002 (schedule generation ≤10 min), and others lack defined test environments, load conditions, or success metrics. Violates IEEE 830 §4.4 (Verification).	IEEE 830 §4.4; ISO/IEC/IEEE 29148 §4.4.2	Impedes objective test case design, acceptance testing, and SLA validation.
Major	**Incomplete Environmental & Operational Requirements** — Missing explicit requirements for disaster recovery, backup frequency/integrity, scalability thresholds (e.g., concurrent users, data volume growth), and failover behavior.	ISO/IEC/IEEE 29148 §2.5 (Environment); §2.6 (Deployment)	Compromises system resilience, operational continuity, and cloud/hybrid deployment assurance.
Minor	**Lack of User Documentation & Training Requirements** — No requirements specifying user guides, role-based training modules, or onboarding workflows.	ISO/IEC/IEEE 29148 §4.3.5 (User Documentation)	Low technical risk, but impacts usability, adoption, and supportability.
Minor	**Ambiguous Source References** — Traceability tags (e.g., *SRL-R1*, *US-006*) are used but lack contextual metadata (e.g., "SRL-R1 = Student Records Policy v3.2, Section 4.1").	ISO/IEC/IEEE 29148 §4.3.8 (Traceability Detail)	Reduces audit transparency; easily remedied with inline annotations or appendix.
Informational	**Opportunity for Automated V&V Integration** — Suggestion to reference specific tools (e.g., *axe-core* for WCAG, *OpenSSL* CLI for cert validation) in verification procedures.	IEEE 830 Annex B (Best Practices)	Enhances efficiency and repeatability of validation—not a compliance gap.

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3. Structural Conformance Assessment

Per IEEE 830 and ISO/IEC/IEEE 29148 template requirements, the SRS exhibits **strong structural alignment**, but with one critical omission:

Section	Status	Notes
Introduction	✅ Compliant	Includes purpose, scope, definitions, references, overview.
Overall Description	✅ Compliant	Covers product perspective, functions, user classes, operating environment, design/implementation constraints.
System Features / Functional Requirements	✅ Compliant	Well-organized, numbered, with acceptance criteria.
External Interface Requirements	✅ Compliant	Covers UI, hardware, software, communication interfaces.
Non-Functional Requirements	⚠️ **Partially Compliant**	Security, usability, performance addressed—but lacks *quantified metrics* for availability (% uptime), scalability (max users/data), and maintainability (MTTR, modularity).

Section	Status	Notes
Other Requirements	❌ **Non-Compliant (Major)**	**Missing dedicated sections for:** – *Verification & Validation Plan* (per IEEE 830 §4.4 — must be referenced or summarized) – *Appendices* (e.g., traceability matrix, glossary expansion, source policy excerpts) – *User Documentation & Training* (ISO/IEC/IEEE 29148 §4.3.5 — optional but recommended for regulated systems)
Traceability Matrix	❌ **Non-Compliant (Critical)**	Not present as a standalone table or appendix. Required by *both* standards (IEEE 830 §4.3.8; ISO/IEC/IEEE 29148 §4.3.8) for bidirectional traceability.

> 🔑 **Key Finding**: The absence of a formal traceability matrix constitutes a **template-level structural deficiency**, not merely an omission of content. This is a *Critical* conformance gap per standard mandates.

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4. Action Items

#	Action Item	Priority	Owner	Target Resolution
1	Develop and embed a **bidirectional traceability matrix** (RTM) mapping each requirement (FR/PR/NFR) to: stakeholder need, source policy/document, use case, user story, and planned test case ID. Must be machine-readable (e.g., CSV/Excel) and included as Appendix A.	🔴 Critical	SRS Author / BA Lead	Within 5 business days
2	Reconcile and formally define **role-specific data access boundaries**, resolving ambiguity between FP-007 and FR-006. Publish updated RBAC schema with explicit permissions per role (HR, Professor, Admin, Student).	🟠 Major	Security Architect / BA	Within 7 business days
3	Augment all performance and latency requirements (PR-003, FR-002, etc.) with **testable thresholds**, including: environment specs (CPU/RAM/network), load conditions (concurrent users, dataset size), and pass/fail criteria.	🟠 Major	Performance Engineer / QA Lead	Within 5 business days
4	Add dedicated subsections under *Non-Functional Requirements*: **Disaster Recovery (RTO/RPO)**, **Scalability (user/data growth targets)**, **Availability (e.g., 99.5% uptime)**, and **Maintainability (e.g., <30-min hotfix deployment)**.	🟠 Major	DevOps Architect / System Architect	Within 7 business days
5	Insert **User Documentation & Training Requirements** (e.g., "FR-DOC-001: System shall provide role-specific interactive onboarding wizards") and update traceability.	🟡 Minor	UX Writer / Product Owner	Within 3 business days
6	Expand traceability tag definitions in an **Appendix B: Source Reference Glossary**, linking each tag (e.g., *US-006*) to its origin document, version, and section.	🟡 Minor	Business Analyst	Within 3 business days

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5. Overall Status

✅ **SRS Validation Status:** `PASS_WITH_ISSUES`

- Meets >90% of structural and content requirements.
- All **Critical** and **Major** issues must be resolved *prior to baselining* for development or regulatory submission.
- **Minor** and **Informational** items are recommended for inclusion in the next revision cycle (v1.2).

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6. Recommendations for Next Steps

1. **Immediate Revision Cycle**: Address Critical & Major action items (Items #1–#4 above) and re-submit for *validation sign-off*.
2. **Cross-Functional Alignment Workshop**: Convene Security, DevOps, QA, and UX teams to jointly validate RBAC logic, DR strategy, and V&V test plans.
3. **Feasibility Prototyping**: Initiate time-boxed (2-week) proof-of-concept for PR-003 (real-time GPS routing) and FR-001 (weather-integrated accessibility routing) to de-risk high-assumption requirements.
4. **Traceability Automation**: Integrate RTM maintenance into the requirements management tool (e.g., Jama, DOORS, or Jira + Xray) to ensure live bidirectional linkage.
5. **Regulatory Readiness Review**: Engage Privacy Officer and Accessibility Coordinator to verify ADA/FERPA alignment post-revision.

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End of Validation Report

Overall Status

pass_with_issues

Summary

Validation report for SRS software_requirement_specs_Archivist_v1.1. Status: pass_with_issues.